ECON251
INDUSTRY AND TRADE IN ASIA
Data Analysis Set A
Report template

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- Website : The world Bank Databank – World Development indicators
- https://databank.worldbank.org/source/world-development-indicators
- Time period (25 years) : ***year 1999 to year 2023***
- Country : Vietnam, Japan, China

Objective

To access data from World Bank, process them to compare the country's development with the more established economies : Japan and China

Question 1 : Data Collection and comparison (30 marks)

Key data **Using the data extracted from the World Bank, World Development Indicators for all THREE countries/economies.** *All values to be presented in 2 decimal places where applicable]* *All computations of should be performed within the excel sheet using appropriate formula or functions*	**Assigned Economy (Vietnam)**	**Japan**	**China**
1.1 Production (4 marks)			
a. Average real GDP (in 2015 constant USD)	$205408091347.90	$4295663922190.04	$8956430058510.33
b. Average real GDP per capita (in 2015 constant USD)	$2257.40	$33802.16	$6534.48
c. Average Economic growth rates (in %)	6.24%	0.69%	8.30%
d. Average trade openness (in %)	140.81%	30.55%	44.59%

1.2 Economic Sectors (9 marks) In this discussion, we only consider the two sectors : primary and secondary sectors . And in the secondary/industry sector , we further investigate the size of the manufacturing sector	**Assigned Economy**	**Japan**	**China**
a. Average share of primary sector to its GDP (using % of GDP)	17.16%	1.14%	9.77%
b. Average share of employment in the primary sector (in %)	47.59%	3.88%	35.72%
c. Average share of industry sector to its GDP (using % of GDP)	36.75%	29.03%	42.62%
d. Average Employment share in the industry sector	22.03%	26.70%	27.60%
e. Share of Manufacturing to its GDP (using % of GDP)	Based on the data: In 1999 (first year, the manufacturing sector contributed approximately 17.41% to GDP. In 2023 (final year), this share increased to 24.18%. The highest value was 24.78%, while the lowest was 17.13%, showing a decline of over 7 percentage points. The average share over 25	Based on the data: In 1999 (first year, the manufacturing sector contributed approximately 22.40% to GDP. In 2023 (final year), this share decreased to 20.58%. The highest value was 22.45%, while the lowest was 19.05%, showing a decline of over 3 percentage points. The average	Based on the data: In 1999 (first year, the manufacturing sector contributed approximately 31.74% to GDP. In 2023 (final year), this share decreased to 25.50%. The highest value was 31.99%, while the lowest was 25.50%, showing a decline of over 6 percentage points. The average share over 25

	years is approximately 20.74%.	share over 25 years is approximately 20.64%.	years is approximately 29.75%.
f. Manufacturing output (in 2015 constant USD)	Based on the data: In 1999 (first year), manufacturing output was USD $15246788409.68. Over time, the output has increased, reaching USD $94969636801.32 in 2023 (final year). The average output over the 25-year period is approximately USD $46893997477.14.	Based on the data: In 1999 (first year), manufacturing output was USD $74315357153 7.73. Over time, the output has increased, reaching USD $10062487538 07.44 in 2023 (final year). The average output over the 25-year period is approximately USD $87350002945 3.17.	Based on the data: In 1999 (first year), manufacturing output was USD $82578045573 1.56. Over time, the output has increased, reaching USD $44907924899 48.81 in 2023 (final year). The average output over the 25-year period is approximately USD $25621786723 50.10.

1.3 Contributors of growth In this section, we will examine the key inputs of production (labour, physical capital, technology, human capital)			
1.3.1 Labour (L) (6 marks)	**Assigned Economy**	**Japan**	**China**
a. Average size of labour force	49465789.52	67095282.92	761162951.60
b. Average unemployment rate	1.82%	3.92%	4.43%
c. Average number of employed	48583835.91	64469917.14	727326152.36
d. Average level of Labour productivity	4078.22	66623.85	12187.14
Key formulas: • Employment rate = 100 – Unemployment Rate • Total employed = Employment rate X labour force • Labour productivity = Real GDP/Total employed			
1.3.2 Capital Goods (K) (3 marks)	**Assigned Economy**	**Japan**	**China**
a. Average level of gross fixed capital formation (K) (in USD)	$62030541787.87	$1110110335810.73	$4694699581163.45 (2015)
b. Average gross fixed capital accumulation growth rate	8.05%	-0.23%	Over the past 25 years, no data was available.
c. Average capital per worker	1217.99	17217.88	6303.67 (2015)
Key Formula *Capital per worker* *= Total K / Total Employed*			

[For the following inputs , data may not be available for some of the assigned economy, as such proceed with the values for Japan and/or China]	**Assigned Economy**	**Japan**	**China**
1.3.3 Level of technology (A) (2 marks) a. Spending on R&D (% of GDP) b. % of firms spending on R&D	a. 0.42% (2022) b. 0.95% (2023)	a. 3.41% (2022) b. Over the past 25 years, no data was available.	a. 2.56% (2022) b. 25.38% (2012)
c. Total Factor Productivity 9 (until 2018) **[Extract from https://data360.worldbank.org/en/dataset/WB_ASPD**	No data available for this country	-0.96% (2018)	3.26% (2018)

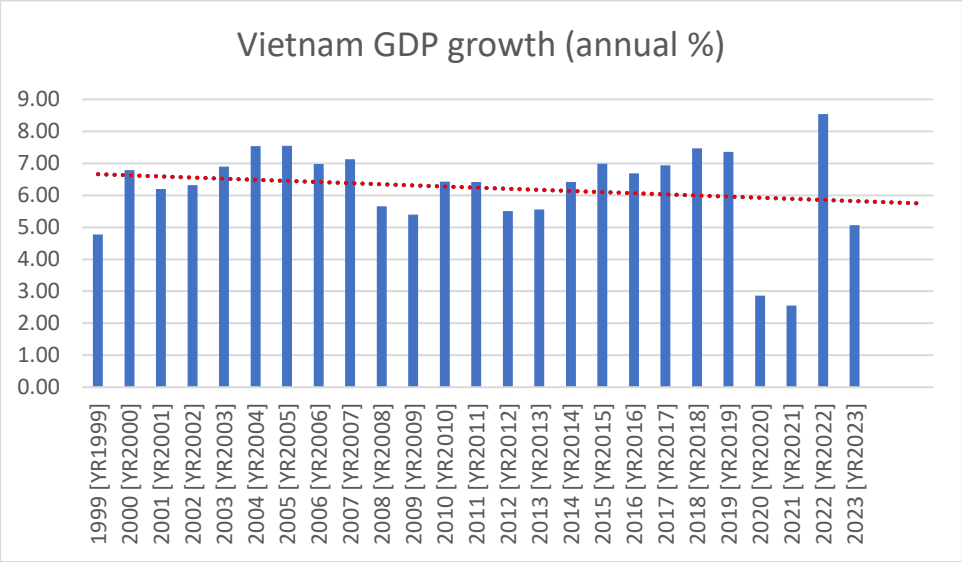
1.3.4 Level of Human Capital (H)(1 marks) a. HDI index	0.69 (2020)	0.80 (2020)	0.65 (2020)

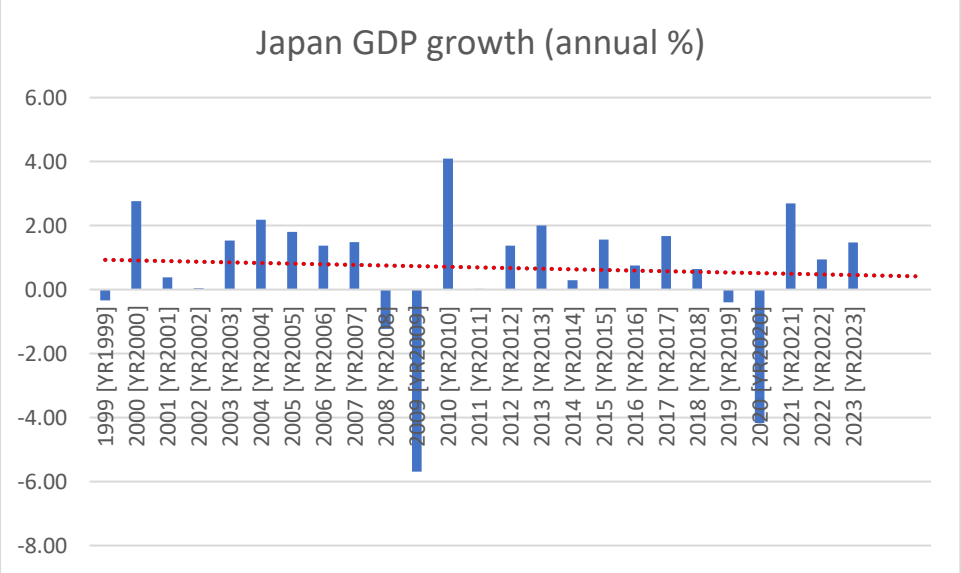
Question 2 Data Visualisation [Total 20 marks]

Step 1: Using suitable excel charts, present the following actual data with each chart containing Title and axis clearly labelled, ***and a trend line*** *to depict the trend for the past 25 years .*

Step 2: Provide a summary (maximum 50 words) for each chart : Highlighting the ranking, the trend line [upward or downward] and the outliers

Chart 1: Economic growth rates of the three economies
[Insert chart below]







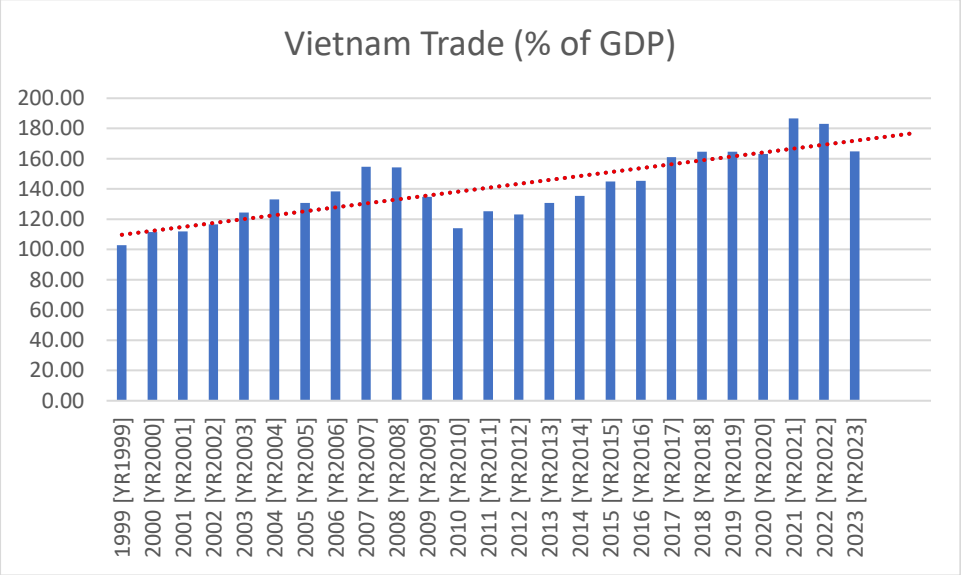
Summary:

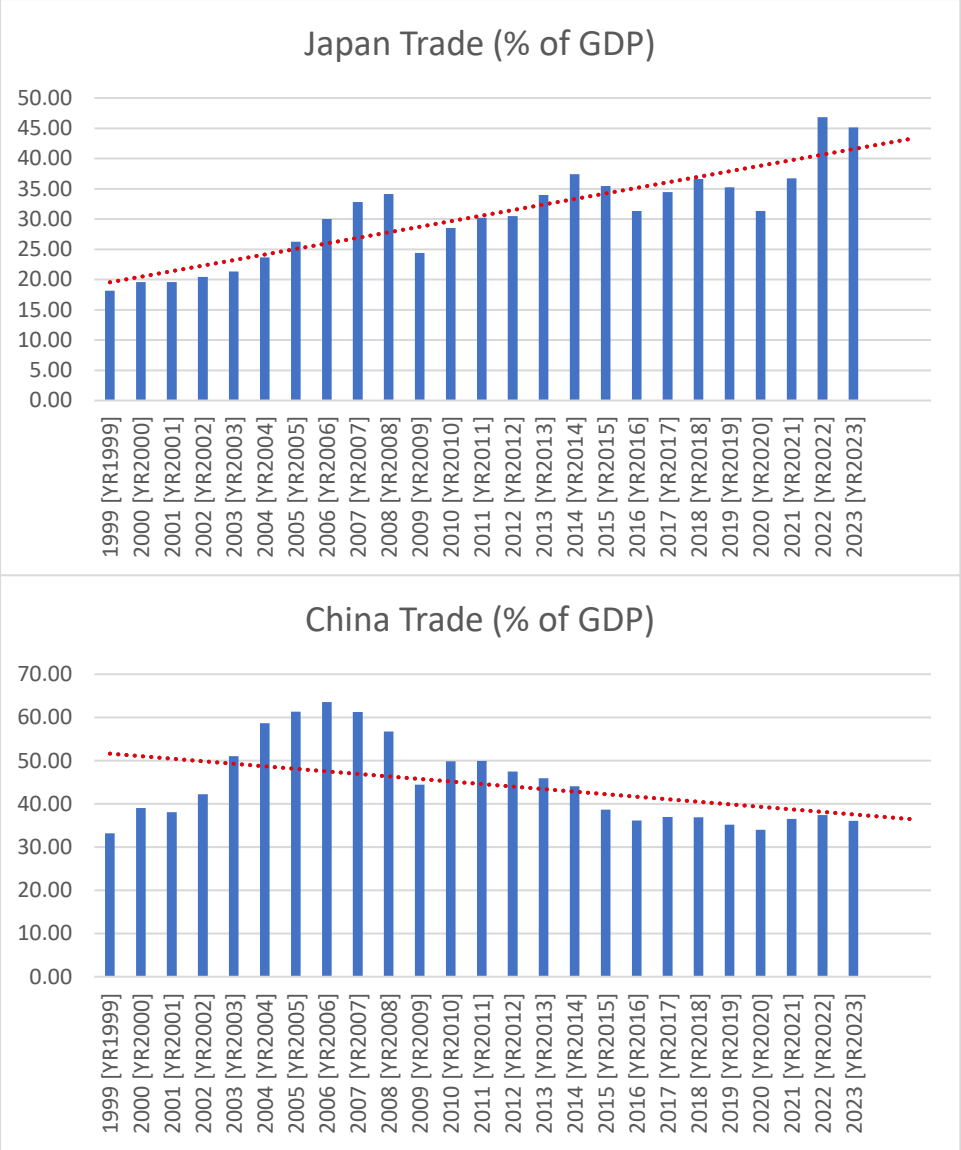
- China ranks first with an average of 8.30% GDP growth, with a downward trend. There is no notable outlier.
- Vietnam ranks second with an average of 6.24% GDP growth, with a slight downward trend. Notable outliers are year 2020 and 2021.
- Japan ranks the third with an average of 0.69% GDP growth, with a slight downward trend. Notable outliers are year 2009 and 2020.

Chart 2 : Trade openness of the three economies

[Please adjust your axis accordingly as it need not start from 0]

[Insert chart below]





Summary:

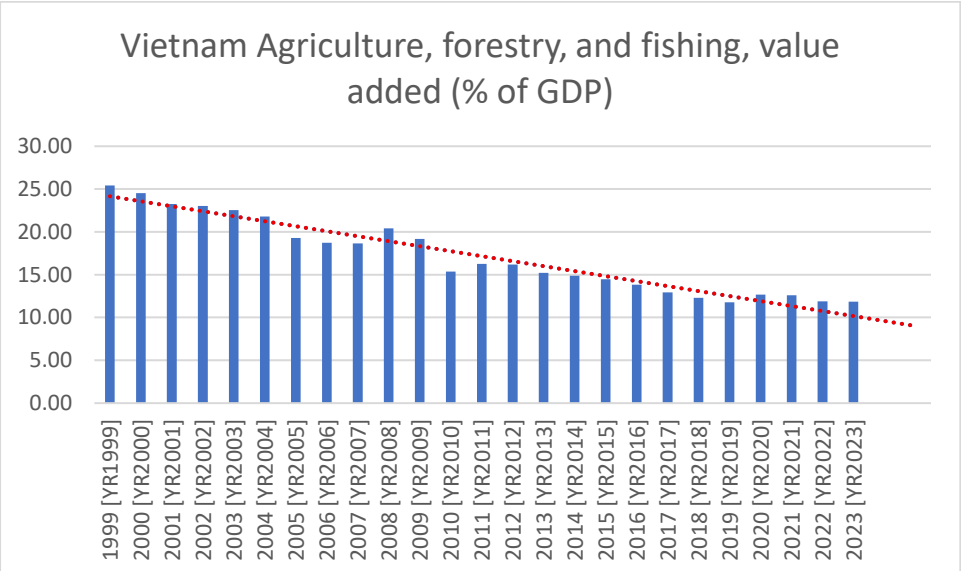
- Vietnam ranks first with an average of 140.81% trade openness, with upward trend. There is no notable outlier.
- China ranks second with an average of 44.59% trade openness, with downward trend. There is no notable outlier.
- Japan ranks third with an average of 30.55% trade openness, with upward trend. There is no notable outlier.

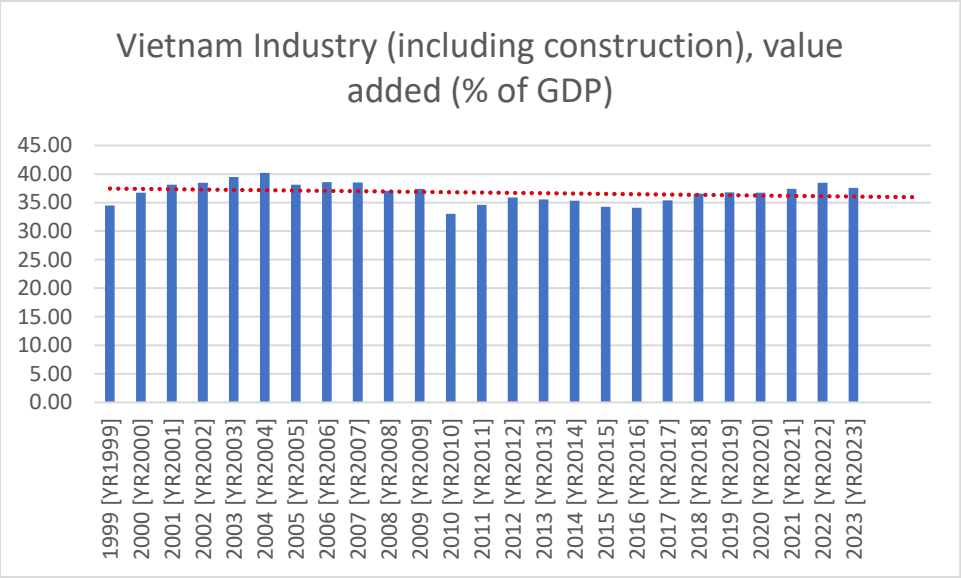
For Charts 3-5 : Present the charts only for the ASSIGNED economy .

Chart 3

The contribution of the two key sectors [i.e. primary and industry] and the employment share in these two sectors

[Insert chart below : There should be 4 graphs]





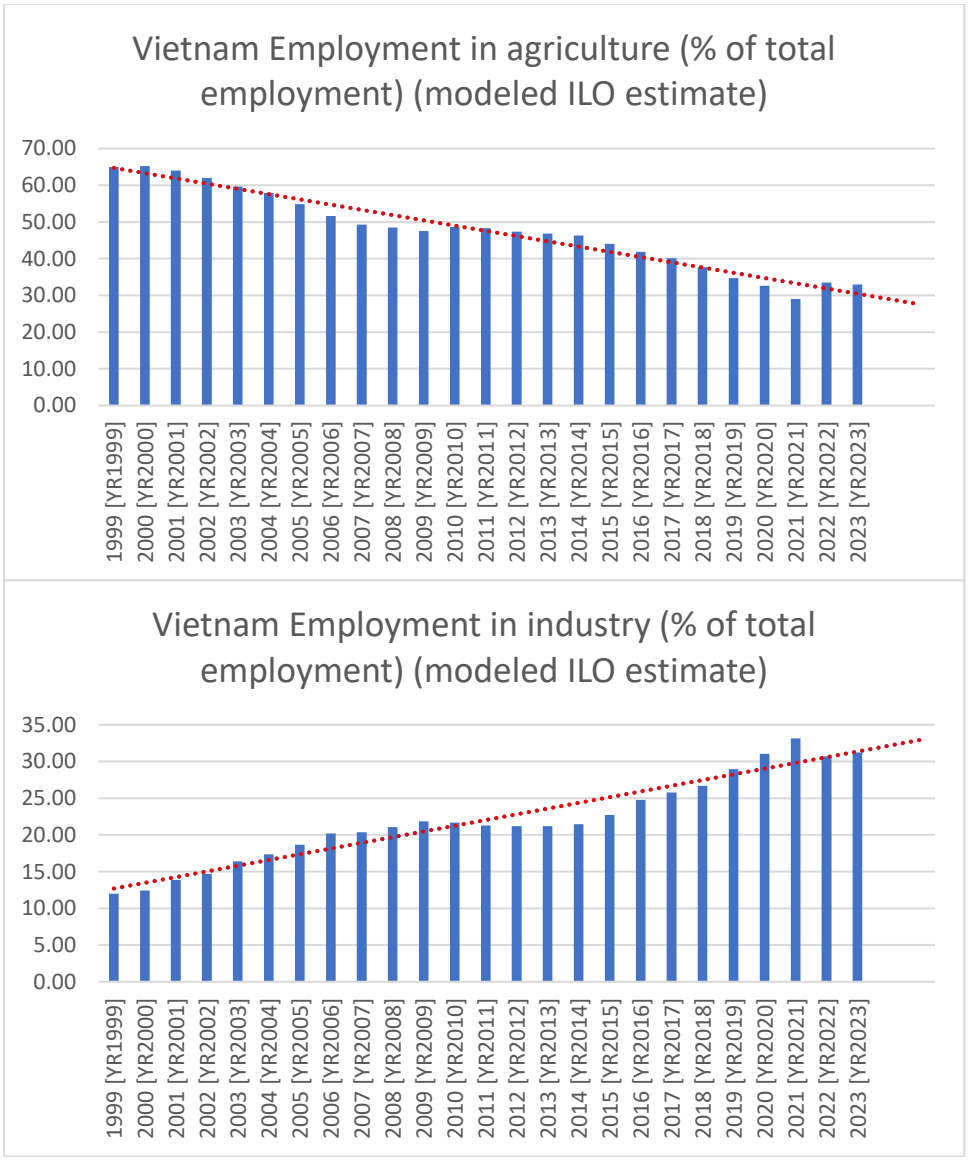


Chart 4
The level of gross fixed capital formation, the level of labour productivity
[Insert chart below]

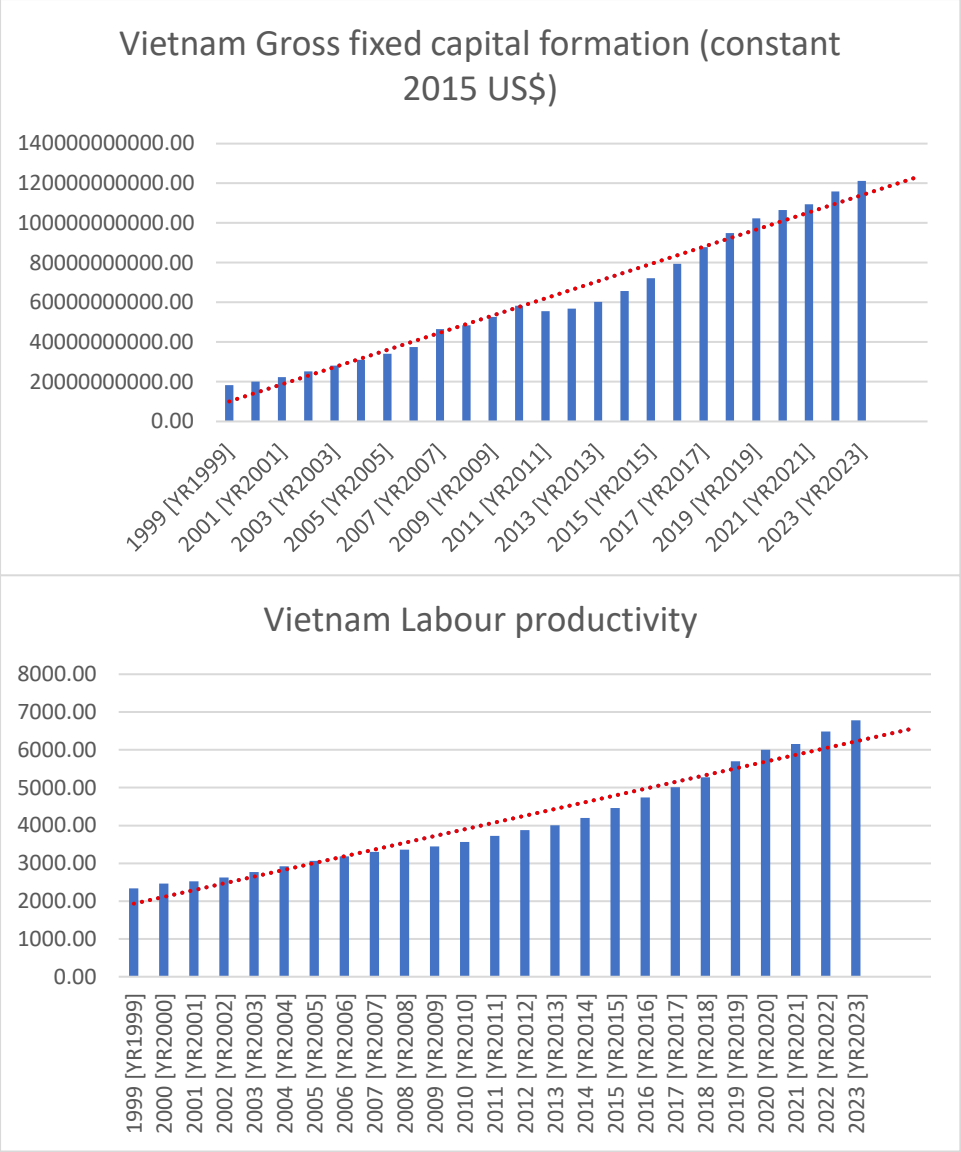
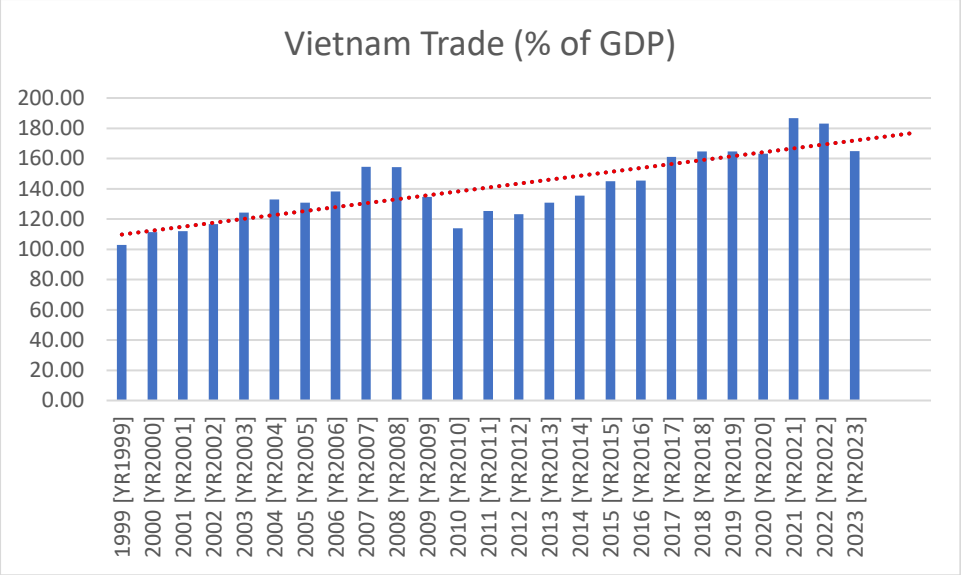


Chart 5
The level of trade openness
[Insert chart below]



Reminder: Check your charts that there should not be any zero value as this will distort the trend line and your observations

Question 3 : Data Analysis [10 marks]

Assessment report on industrialisation of the assigned economy (Max 200 words)

[Hint: Using evidence from charts, correlation computation, to discuss if the economy is anchoring on the secondary sector for its economic growth]

From 1999 to 2023, Vietnam's secondary sector (which includes manufacturing, construction, and industry-related activities) contributed an average of 36.75% to the country's GDP, while the primary sector (mainly agriculture, forestry, and fishing) contributed a lower average of 17.16%.

This indicates that the industrial sector plays a significantly larger and more impactful role in Vietnam's economy, acting as a key engine for growth and development.

Notably, the correlation between the secondary sector's share of GDP and Vietnam's overall GDP growth is 1.00. This means there is a perfect positive linear relationship between the two — as the contribution from the industrial sector increases, Vietnam's GDP also rises in a directly proportional way.

This strong correlation highlights the strategic importance of continued investment in industrialization, manufacturing capabilities, and infrastructure to sustain Vietnam's long-term economic growth.

~End of Assignment ~